

1. Determine the equation of the tangent line to $g(x) = \left(\frac{1}{x^3} + 1\right)(x-1)$ at $x = -1$.

2. If $g(x) = \frac{f(x)}{\sqrt{x-1}}$, where $f(5) = 8$, and $f'(5) = -5$, find $g'(5)$.

3. Find the points on the function $f(x) = \frac{x+9}{x+8}$ where the tangent lines pass through the origin.

4. Find the equation of the normal to the curve $f(x) = \sqrt[3]{x^2 - 1}$ at the point where $x = 3$.

5. Let f and g be functions such that $g(x) = \frac{f(x)}{x}$. If $y = 2x - 3$ is the equation of the tangent to the graph of $f(x)$ at $x = 1$, what is the equation of the line tangent to the graph of $g(x)$ at $x = 1$?

6. Find the points on the curve $f(x) = \frac{x}{x+1}$ where the **normal** line is parallel to $x + y = 2$.

7. Let $f(x) = \sqrt{ax^2 + b}$. Find values of a and b such that the linear equation $7x + 2y = 5$ is tangent to $f(x)$ at $x = -1$.

8. Find the area of the triangle determined by the coordinate axes and the tangent to the curve $xy = 1$ at $x = 1$.

9. Consider the curve $y = a\sqrt{x} + \frac{b}{\sqrt{x}}$ where a and b are constants. The normal to this curve at the point where $x = 4$ is $4x + y = 22$. Find the values of a and b .